AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

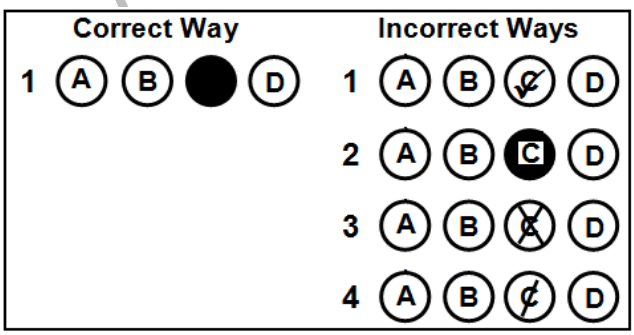
Computer Science Paper I

Time: 40 minutes Marks: 30

AKU-EB May Examination 2017

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. Which of the following statements is TRUE about system software?

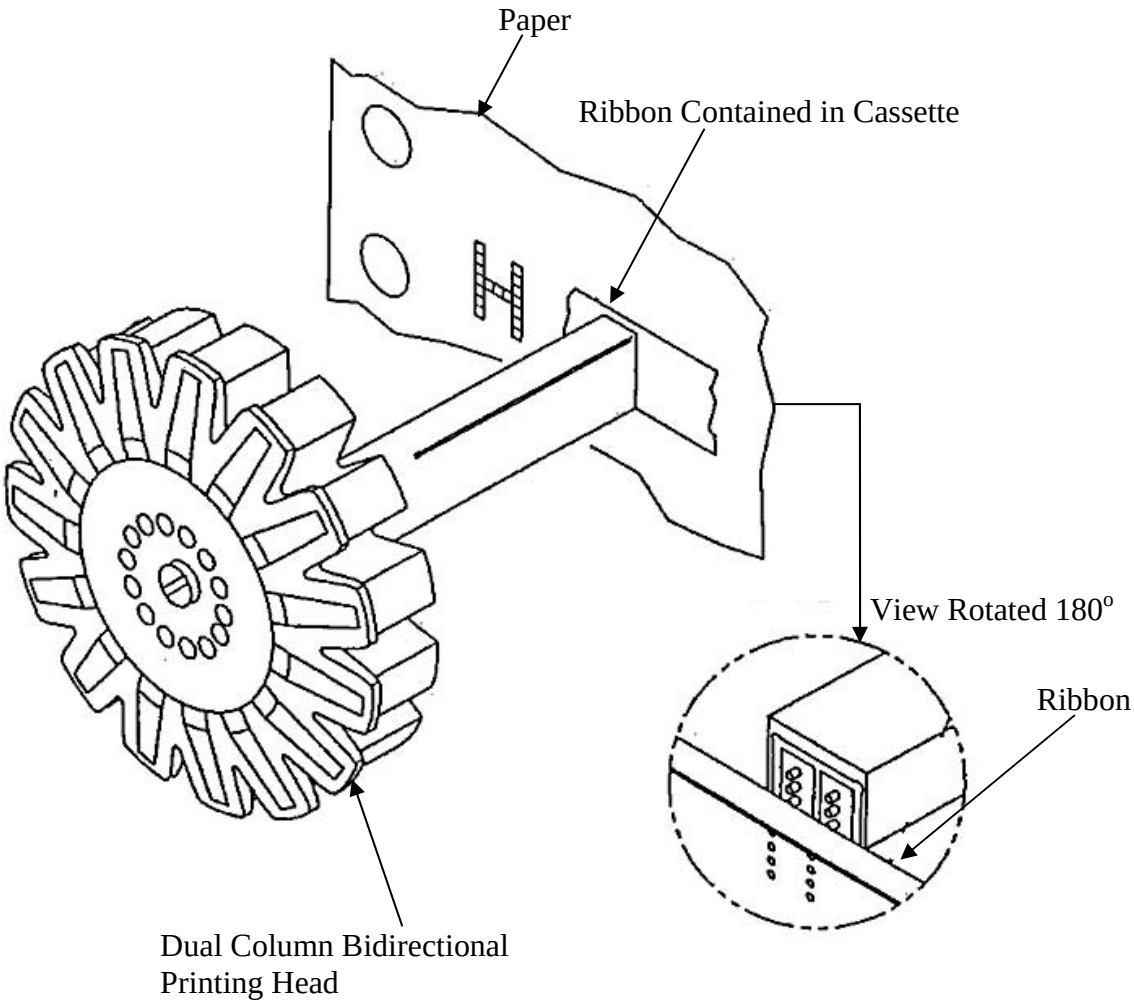
   A. It is general purpose software.
   B. It is specific purpose software.
   C. It is optional for effective operation of computer.
   D. It provides an environment that stops other software from running.

2. All of the following are functions of a scanner EXCEPT to recognise

   A. voices.
   B. pen marks.
   C. magnetic ink.
   D. text characters.

2017

3. The printing mechanism shown in the given picture belongs to



   A. daisy wheel printer.
   B. dot matrix printer.
   C. chain printer.
   D. laser printer.

4. An example of application software is

   A. UNIX.
   B. LINUX.
   C. Microsoft Word.
   D. Microsoft Windows.

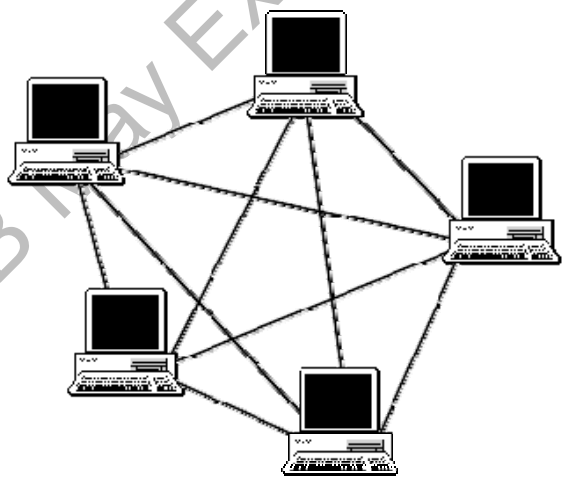
5. If one alphabet of English language occupies eight bits in memory, then the number of **bytes** that will be occupied by the word COMPUTER is

   A. 8 bytes.
   B. 16 bytes.
   C. 32 bytes.
   D. 64 bytes.

6. In which of the following stages of system development life cycle (SDLC) are changes made to software and hardware to support the system's operational effectiveness?

   A. Testing
   B. Debugging
   C. Maintenance
   D. Problem Analysis

7. The given topology is an example of



   A. bus topology.
   B. tree topology.
   C. star topology.
   D. mesh topology.

8. In open system interconnection (OSI) model, the layer responsible for conveying bit streams to the medium is the

   A. physical layer.
   B. network layer.
   C. data-link layer.
   D. transport layer.

AKU-EB May Examination 2017

9. The purpose of the application layer of open system interconnection (OSI) model is that it

   A. ensures error free delivery of messages.
   B. provides logical addressing to data packets.
   C. establishes/ ends connection between two nodes.
   D. provides user interface to access network services.

10. Wide Area Network (WAN) provides connectivity

   A. within a house.
   B. within an office.
   C. between two cities.
   D. between two rooms.

11. The standard designed to be used in Metropolitan Area Networks (MANs) is

   A. TCP/IP
   B. IEEE 802.6
   C. IEEE 802.3
   D. X.25 Protocol

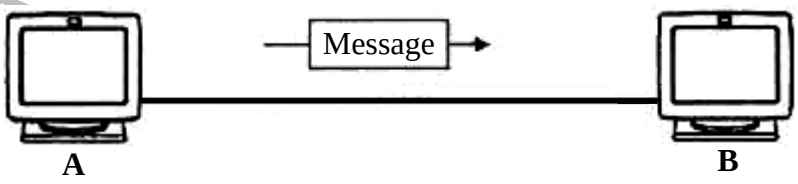
12. All of the following are features of peer-to-peer networking EXCEPT

   A. ease in installation and maintenance.
   B. small group of network users.
   C. decentralisation of resources.
   D. strong security measures.

13. Which of the following can be the mode of communication between a computer and a keyboard?

   A. Serial
   B. Parallel
   C. Simplex
   D. Half duplex

14. Consider the data communication between two computers shown below.



   If computer **A** sends data to computer **B**, then computer **B** is called the

   A. sender.
   B. medium.
   C. receiver.
   D. transmitter.

AKU-EB May Examination 2017

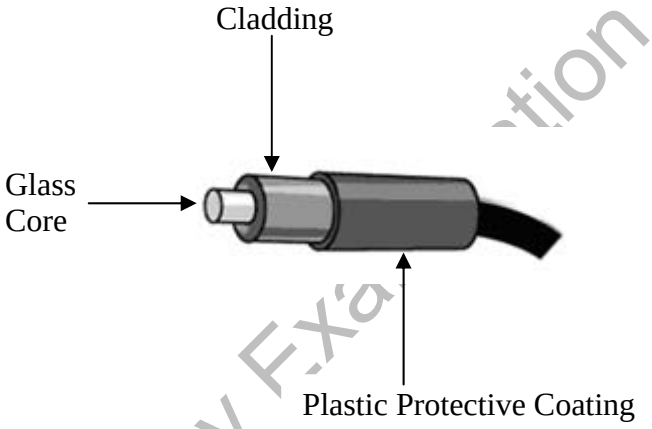
15. Which of the following provides the highest data transfer rate?

A. Microwave
B. Optical fibre
C. Radio waves
D. Coaxial cable

16. The possible number of characters for 7-bit coding scheme is

A. 2^7
B. 7^2
C. $2^6 + 1$
D. $6^2 + 1$

17. The given cable structure represents



A. coaxial cable.
B. fiber-optic cable.
C. twisted-pair cable.
D. standard telephone line.

18. When compared with synchronous transmission, which of the following statements is TRUE about asynchronous transmission?

A. There are no gaps in transmission.
B. It is essential to maintain the timing.
C. Each data group has a start and stop bit.
D. The receiver counts the bits and reconstructs bytes.

19. Which of the following is TRUE about Computer Aided Design (CAD)?

A. It uses computer to build objects.
B. It creates vector-based drawings of objects.
C. It uses machines that are controlled by computers.
D. It uses programmable machinery in the manufacturing process.

AKU-EB May Examination 2017

20. Which of the following is a potential drawback of e-commerce?

A. It allows comparing prices among several providers.
B. It needs a well-built website but not the infrastructure.
C. Consumer is unable to experience the product before purchase.
D. Consumer can read the online good or bad reviews about a product.

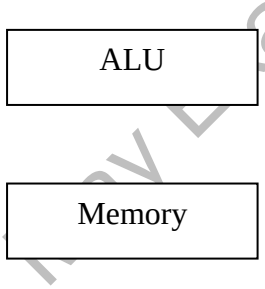
21. A graphic package can do all of the following EXCEPT

A. edit images.
B. insert images.
C. search images.
D. manipulate images.

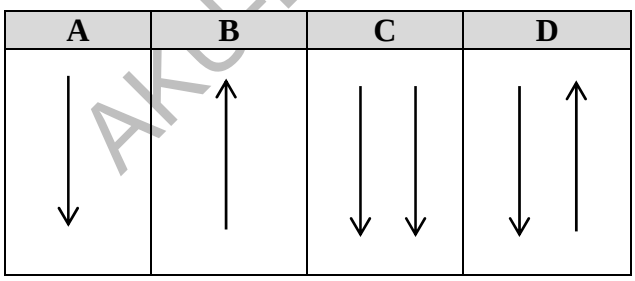
22. Which of the following is a potential disadvantage of e-banking?

A. It is not constrained by timings.
B. It provides an easy way to make payments.
C. It requires the internet for any activity to be done.
D. It allows to do transactions from anywhere in the world.

23. Consider the given part of computer block diagram containing arithmetic logic unit (ALU) and memory.



These two units of computer are connected via bus(es). To connect these two units, the direction of bus(es) will be



24. The unit of computer system that calculates the square root of any number is

A. input unit.
B. output unit.
C. control unit.
D. arithmetic logic unit.

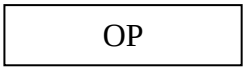
AKU-EB May Examination 2017

25. Which of the following is the CORRECT format of instruction code?

A. 

B. 

C. 

D. 

26. Which of the following ensures the receiver that the message is coming from the intended sender, not a hacker?

A. Confidentiality
B. Authentication
C. Decryption
D. Encryption

27. Which of the following is a valid feature of Windows operating system?

A. Compiler
B. Clipboard
C. Interpreter
D. Assembler

28. Consider the text written in box **A** and box **B**.

A

Computer is an electronic device that takes input, processes it and gives the output.

B

Computer is an electronic device that takes input, processes it and gives the output.

Which of the following text alignments will align the text in box **A** in the same way as the text in box **B**?

A. Left
B. Right
C. Center
D. Justify

AK – May Examination 2017

29. Consider the part of MS Excel sheet shown below.

	A	B	C
1	5	4	1
2	10	6	3
3	15	8	5
4	20	10	7
5	25	12	9
6	30	14	11

Which of the following MS Excel formulae will **add** the values of cells A1, A2, A3, A4, A5 and B2 giving result of 81?

A. =ADD(A1:A5:B2)
B. =ADD(A1:A5,B2)
C. =SUM(A1:A5:B2)
D. =SUM(A1:A5,B2)

30. Which of the following options represents the hostname in the given email address?

alizeh@spark.berkeley.edu

A. edu
B. spark
C. alizeh
D. berkeley

AKU-EB May Examination 2017

END OF PAPER